| | (a) failed lifts (%), change | | | | | | (b) grip (% of weight), change | | | | | | (c) episode time (s), change | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | B0 | B0S | B2 | B4g | B4t | B5 | B0 | B0S | B2 | B4g | B4t | B5 | B0 | B0S | B2 | B4g | B4t | B5 |
| Reference | 0.0 | 0.0 | 11.5 | 0.0 | 0.0 | 0.0 | 101 | 101 | 128 | 136 | 128 | 136 | 6.03 | 6.03 | 5.40 | 5.40 | 5.45 | 5.40 |
| E1 heavier objects 0.2-0.5 kg | 0 (0.0) | 0 (0.0) | **+14.5*** (26.0) | 0 (0.0) | 0 (0.0) | 0 (0.0) | +1 (102) | +1 (102) | +2 (129) | +3 (139) | +2 (129) | +2 (138) | -0.05 (5.98) | -0.05 (5.98) | -0.05 (5.35) | -0.05 (5.35) | -0.04 (5.41) | -0.05 (5.35) |
| E2 slippery objects mu 0.2-0.4 | 0 (0.0) | 0 (0.0) | **-8.1*** (3.4) | 0 (0.0) | 0 (0.0) | 0 (0.0) | **+2008*** (2109) | **+2008*** (2109) | **+1920*** (2047) | **+1991*** (2127) | **+1995*** (2123) | **+1990*** (2126) | **-2.32*** (3.71) | **-2.32*** (3.71) | **-2.26*** (3.14) | **-1.89*** (3.51) | **-1.92*** (3.53) | **-1.89*** (3.51) |
| E3 slip-monitor noise 1.5 mm | 0 (0.0) | 0 (0.0) | **-11.5*** (0.0) | 0 (0.0) | 0 (0.0) | 0 (0.0) | **+3007*** (3108) | **+3007*** (3108) | **+2867*** (2994) | **+2973*** (3108) | **+2981*** (3108) | **+2972*** (3108) | **-2.63*** (3.40) | **-2.63*** (3.40) | **-2.55*** (2.85) | **-2.00*** (3.40) | **-2.05*** (3.40) | **-2.00*** (3.40) |
| E4 weight error +-25% | 0 (0.0) | 0 (0.0) | -2.6 (8.9) | 0 (0.0) | 0 (0.0) | 0 (0.0) | +1 (102) | +1 (102) | +1 (129) | +1 (137) | +1 (129) | +1 (137) | -0.02 (6.01) | -0.02 (6.01) | -0.02 (5.38) | -0.02 (5.38) | -0.02 (5.43) | -0.02 (5.38) |
| E5 coarser probe step 0.8 | 0 (0.0) | 0 (0.0) | -4.9 (6.6) | 0 (0.0) | 0 (0.0) | 0 (0.0) | **+5*** (106) | **+5*** (106) | **+7*** (135) | **+7*** (142) | **+7*** (135) | **+6*** (142) | **-1.26*** (4.77) | **-1.26*** (4.77) | **-1.26*** (4.14) | **-1.26*** (4.14) | **-1.26*** (4.19) | **-1.26*** (4.14) |
| E6 finer probe step 0.95 | 0 (0.0) | 0 (0.0) | **+18.5*** (30.0) | 0 (0.0) | 0 (0.0) | 0 (0.0) | **-3*** (98) | **-3*** (98) | **-5*** (123) | **-3*** (132) | **-5*** (123) | **-4*** (133) | **+2.56*** (8.59) | **+2.56*** (8.59) | **+2.56*** (7.96) | **+2.56*** (7.96) | **+2.57*** (8.02) | **+2.56*** (7.96) |
| E7 lift height 20 cm | 0 (0.0) | 0 (0.0) | **+46.4*** (57.9) | **+37.8*** (37.8) | **+4.1*** (4.1) | **+38.6*** (38.6) | **+6*** (106) | **+6*** (106) | **+38*** (166) | **+41*** (176) | **+38*** (166) | **+41*** (177) | 0 (6.03) | 0 (6.03) | 0 (5.40) | 0 (5.40) | **-0.02*** (5.43) | 0 (5.40) |
| E8 softer pads solref 0.006 | 0 (0.0) | 0 (0.0) | +3.5 (15.0) | 0 (0.0) | 0 (0.0) | 0 (0.0) | **-3*** (98) | **-3*** (98) | **-4*** (123) | **-4*** (132) | **-4*** (123) | **-4*** (132) | **+0.12*** (6.15) | **+0.12*** (6.15) | **+0.12*** (5.52) | **+0.12*** (5.52) | **+0.12*** (5.57) | **+0.12*** (5.52) |
| E9 earlier kinetic switch 5 mm/s | 0 (0.0) | 0 (0.0) | **+54.9*** (66.4) | **+46.9*** (46.9) | **+53.9*** (53.9) | **+46.9*** (46.9) | +0.10 (101) | +0.10 (101) | +0.13 (128) | +0.13 (136) | +0.13 (128) | +0.13 (136) | -0.005 (6.03) | -0.005 (6.03) | -0.005 (5.40) | -0.005 (5.40) | -0.01 (5.44) | -0.005 (5.40) |
| S1 shield r_max = 1.0 | 0 (0.0) | 0 (0.0) | 0 (11.5) | **+5.0*** (5.0) | **+5.0*** (5.0) | **+7.5*** (7.5) | 0 (101) | 0 (101) | 0 (128) | **-7*** (129) | 0 (128) | **-9*** (127) | 0 (6.03) | 0 (6.03) | 0 (5.40) | 0 (5.40) | **-0.04*** (5.41) | 0 (5.40) |
| S2 shield r_max = 1.1 | 0 (0.0) | 0 (0.0) | 0 (11.5) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 0 (101) | 0 (101) | 0 (128) | **-0.16*** (136) | 0 (128) | **-0.19*** (136) | 0 (6.03) | 0 (6.03) | 0 (5.40) | 0 (5.40) | **-0.000*** (5.45) | 0 (5.40) |
| S3 shield r_max = 1.4 | 0 (0.0) | 0 (0.0) | 0 (11.5) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 0 (101) | 0 (101) | 0 (128) | 0 (136) | 0 (128) | 0 (136) | 0 (6.03) | 0 (6.03) | 0 (5.40) | 0 (5.40) | 0 (5.45) | 0 (5.40) |
| S4 slip tolerance 2 mm | 0 (0.0) | 0 (0.0) | **+2.9*** (14.4) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 0 (101) | 0 (101) | 0 (128) | 0 (136) | 0 (128) | 0 (136) | 0 (6.03) | 0 (6.03) | 0 (5.40) | 0 (5.40) | 0 (5.45) | 0 (5.40) |
| S5 slip tolerance 10 mm | 0 (0.0) | 0 (0.0) | **-3.0*** (8.5) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 0 (101) | 0 (101) | 0 (128) | 0 (136) | 0 (128) | 0 (136) | 0 (6.03) | 0 (6.03) | 0 (5.40) | 0 (5.40) | 0 (5.45) | 0 (5.40) |
| S6 no sensing margin | 0 (0.0) | 0 (0.0) | 0 (11.5) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 0 (101) | 0 (101) | 0 (128) | 0 (136) | 0 (128) | 0 (136) | 0 (6.03) | 0 (6.03) | 0 (5.40) | 0 (5.40) | 0 (5.45) | 0 (5.40) |
| T1 trained on r = 1.0 only | 0 (0.0) | 0 (0.0) | **+12.9*** (24.4) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 0 (101) | 0 (101) | **-2*** (125) | -0.02 (136) | **-2*** (125) | 0 (136) | 0 (6.03) | 0 (6.03) | **-0.000*** (5.40) | **-0.000*** (5.40) | **+0.02*** (5.47) | 0 (5.40) |
| T2 wider time action (scale 1.5) | 0 (0.0) | 0 (0.0) | **+30.8*** (42.3) | **+23.8*** (23.8) | **+13.8*** (13.8) | **+23.8*** (23.8) | 0 (101) | 0 (101) | **+47*** (175) | **+49*** (185) | **+47*** (175) | **+49*** (185) | 0 (6.03) | 0 (6.03) | **-0.12*** (5.28) | **-0.12*** (5.28) | **-0.15*** (5.30) | **-0.12*** (5.28) |
| T3 grip weight lambda = 10 | 0 (0.0) | 0 (0.0) | **-6.6*** (4.9) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 0 (101) | 0 (101) | **-2*** (125) | **-6*** (130) | **-2*** (125) | **-18*** (118) | 0 (6.03) | 0 (6.03) | **+0.03*** (5.43) | **+0.03*** (5.43) | **+0.01*** (5.46) | **+0.13*** (5.53) |
| T4 double training budget | 0 (0.0) | 0 (0.0) | **-11.0*** (0.5) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 0 (101) | 0 (101) | **+4*** (131) | **+1*** (137) | **+4*** (131) | **+1*** (137) | 0 (6.03) | 0 (6.03) | **-0.003*** (5.40) | **-0.003*** (5.40) | **-0.02*** (5.42) | **-0.003*** (5.40) |
| R0L reference, lag-free switch | 0 (0.0) | 0 (0.0) | 0 (11.5) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 0 (101) | 0 (101) | 0 (128) | 0 (136) | 0 (128) | 0 (136) | 0 (6.03) | 0 (6.03) | 0 (5.40) | 0 (5.40) | 0 (5.45) | 0 (5.40) |
| T2L wider time action, lag-free switch | 0 (0.0) | 0 (0.0) | **-9.1*** (2.4) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 0 (101) | 0 (101) | **+47*** (175) | **+49*** (185) | **+47*** (175) | **+49*** (185) | 0 (6.03) | 0 (6.03) | **-0.12*** (5.28) | **-0.12*** (5.28) | **-0.15*** (5.30) | **-0.12*** (5.28) |
| E7L lift height 20 cm, lag-free switch | 0 (0.0) | 0 (0.0) | **-2.9*** (8.6) | 0 (0.0) | 0 (0.0) | 0 (0.0) | **+6*** (106) | **+6*** (106) | **+38*** (166) | **+41*** (176) | **+38*** (166) | **+41*** (177) | 0 (6.03) | 0 (6.03) | 0 (5.40) | 0 (5.40) | **-0.02*** (5.43) | 0 (5.40) |
| E9L switch 5 mm/s, lag-free switch | 0 (0.0) | 0 (0.0) | **+54.4*** (65.9) | **+45.5*** (45.5) | **+23.4*** (23.4) | **+45.8*** (45.8) | +0.10 (101) | +0.10 (101) | +0.13 (128) | +0.13 (136) | +0.13 (128) | +0.13 (136) | -0.005 (6.03) | -0.005 (6.03) | -0.005 (5.40) | -0.005 (5.40) | -0.01 (5.44) | -0.005 (5.40) |